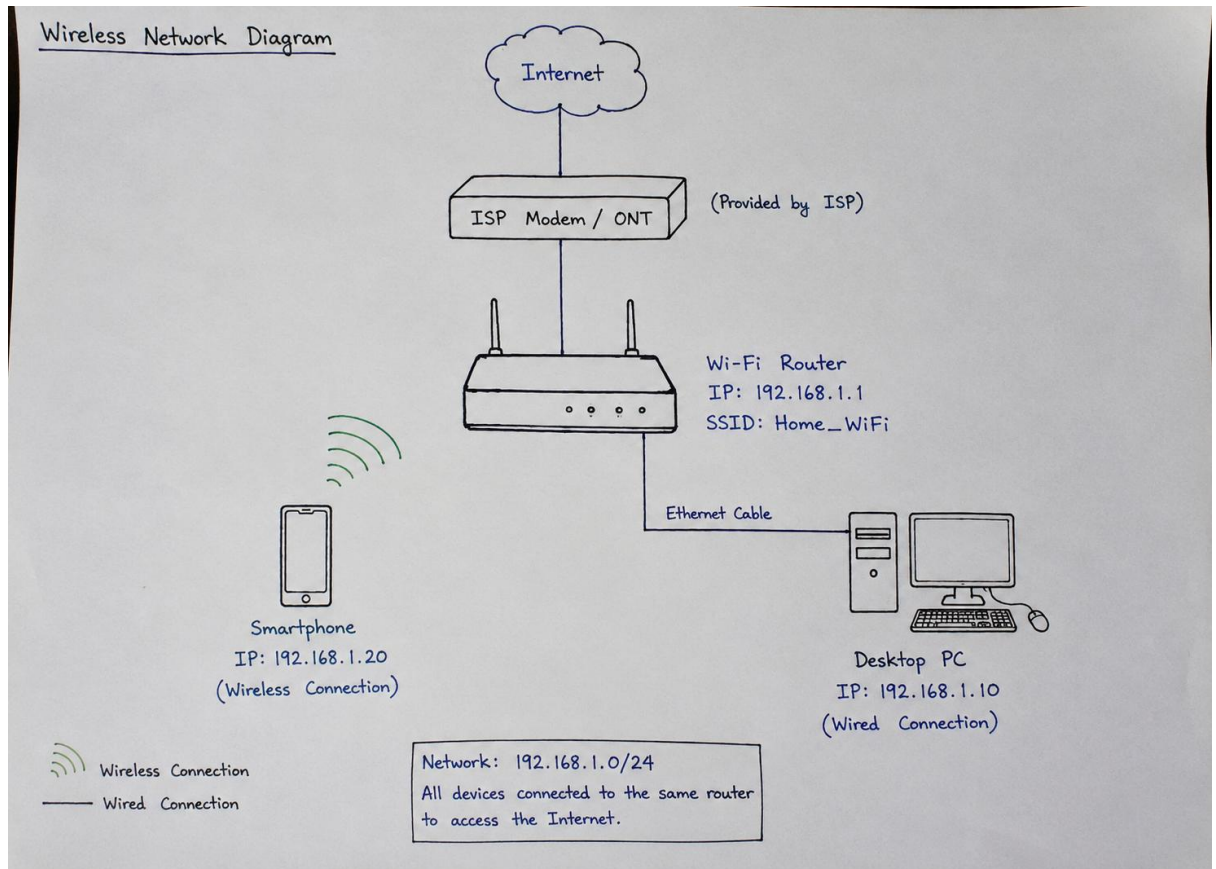


Wireless Networking

1) List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Wi-Fi Standard	Frequency Band	Maximum Speed	Key Difference
802.11n (Wi-Fi 4)	2.4 GHz & 5 GHz	Up to 600 Mbps	First standard to support both 2.4 GHz and 5 GHz with MIMO technology.
802.11ac (Wi-Fi 5)	5 GHz	Up to 3.5 Gbps	Faster than 802.11n and uses only the 5 GHz band.
802.11ax (Wi-Fi 6)	2.4 GHz & 5 GHz (6 GHz with Wi-Fi 6E)	Up to 9.6 Gbps	Offers higher speed, lower latency, and better performance for many connected devices.

- 2) Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.



Labels

- **Wireless Connection:** Smartphone ↔ Wi-Fi Router
- **Wired Connection:** Desktop PC ↔ Router (Ethernet Cable)
- **Router IP:** 192.168.1.1
- **SSID:** Home_WiFi

- 3) Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Hint: Consider both signal strength and security when choosing.

SSID	Signal Strength	Security Type
Home_WiFi	Excellent (95%)	WPA3
Neighbor_WiFi	Good (70%)	WPA2
Cafe_FreeWiFi	Fair (45%)	Open (No Security)

I would choose **Home_WiFi** because:

- It has the **strongest signal (95%)**.
- It uses **WPA3**, which provides the highest level of wireless security.
- It offers better speed, stability, and protection against unauthorized access.

- 4) Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

SSID hiding is a feature where a Wi-Fi router does not broadcast the network name (SSID). As a result, the network does not appear in the list of available Wi-Fi networks, and users must enter the SSID manually to connect.

Advantage

- It makes the network less visible to casual users, reducing accidental connection attempts.

Disadvantage

- It does **not** provide strong security because attackers can still detect a hidden SSID using network analysis tools. It can also make it less convenient for legitimate users to connect.

Conclusion: Hiding the SSID should not be relied on as a security measure.

Using **WPA3 (or WPA2 if WPA3 is unavailable)** with a strong password provides much better protection.